SRMJEEE 2026 — Official Syllabus

SRM Joint Engineering Entrance Examination | B.Tech (UG) | Remote Proctored Online Mode

Phase	Exam Dates	Application Deadline
Phase 1	24 Apr – 29 Apr 2026	16 Apr 2026
Phase 2	10 Jun – 15 Jun 2026	04 Jun 2026
Phase 3	04 Jul – 05 Jul 2026	30 Jun 2026

** Dates are tentative. Exam conducted via Remote Proctored Online Mode from home (Desktop/Laptop).*

■ PHYSICS — 10 Units · 35 Questions · +1/0 Marking

Unit 1: Units, Measurement & Laws of Motion	S.I. units, Errors, Dimensional analysis, Newton's laws, Projectile motion, Circular motion, Work-Energy theorem, Elastic & inelastic collisions
Unit 2: Gravitation, Solids & Fluids	Universal gravitation, Escape velocity, Geostationary satellites, Hooke's law, Elasticity (Young's, Bulk, Shear moduli), Surface tension, Viscosity, Bernoulli's theorem
Unit 3: Electrostatics	Coulomb's law, Superposition principle, Electric field, Gauss's theorem, Electric dipole, Capacitors (series/parallel), Dielectrics, Energy stored in capacitor
Unit 4: Current Electricity	Ohm's law, V-I characteristics, Kirchhoff's laws, Wheatstone bridge, Metre bridge, Potentiometer, EMF, Thermoelectric current
Unit 5: Magnetism & Magnetic Effects of Current	Earth's magnetic field, Tangent galvanometer, Deflection magnetometer, Biot-Savart's law, Moving coil galvanometer, Ampere's law, Conversion to ammeter/voltmeter
Unit 6: EMI, AC & EM Waves	Faraday's laws, Lenz's law, Eddy currents, Self & mutual induction, LCR series circuit, Resonance, AC generator, Transformer, EM spectrum
Unit 7: Optics	Reflection, Refraction, TIR, Optical fibers, Lenses, Prism, Young's double slit, Huygens' principle, Diffraction (single slit), Polarisation
Unit 8: Dual Nature & Atomic Physics	Photoelectric effect, de Broglie relation, Hertz & Lenard's observations, Rutherford's model, Bohr model, Hydrogen spectrum
Unit 9: Nuclear Physics	Binding energy, Mass defect, Bainbridge spectrometer, Radioactivity, Half-life, Carbon dating, Nuclear fission & fusion, Cosmic rays
Unit 10: Electronic Devices	P-N junction diode, Biasing, Transistors, Logic gates (NOT/OR/AND/NOR/NAND) as universal gates, De Morgan's theorem

■ CHEMISTRY — 12 Units · 35 Questions · +1/0 Marking

Unit 1: Solutions	Types of solutions, Concentration (molality, molarity, mole fraction), Raoult's law, Colligative properties, Osmotic pressure, van't Hoff factor
Unit 2: Electrochemistry	Redox reactions, Conductance, Kohlrausch's law, Galvanic & electrolytic cells, Nernst equation, Gibbs energy change, Corrosion
Unit 3: Chemical Kinetics	Rate of reaction, Order, Molecularity, Half-life (zero & first order), Arrhenius equation, Activation energy, Collision theory
Unit 4: Surface Chemistry	Adsorption (physisorption/chemisorption), Catalysis, Colloids, Tyndall effect, Brownian motion, Electrophoresis, Coagulation
Unit 5: p-Block Elements	Group 16 (O_2, O_3, H_2SO_4, SO_2), Group 17 (Halogens, HCl, Interhalogen compounds), Group 18 (Noble gases, Xenon fluorides)
Unit 6: d & f-Block Elements	Transition metals, Electronic configuration, Oxidation states, Colour, Magnetic properties, Catalytic property, $K_2Cr_2O_7$, $KMnO_4$

Unit 7: Coordination Compounds	Ligands, Coordination number, IUPAC nomenclature, Werner's theory, VBT, CFT, Geometrical & ionization isomerism
Unit 8: Haloalkanes & Haloarenes	C-X bond nature, Substitution mechanisms, Optical rotation, Environmental effects of chloroform, freons, DDT
Unit 9: Alcohols, Phenols & Ethers	Nomenclature, Preparation of primary alcohols, Dehydration mechanism, Acidic nature of phenol, Electrophilic substitution, Ethers
Unit 10: Aldehydes, Ketones & Carboxylic Acids	Carbonyl group, Nucleophilic addition, Aldol condensation, Cannizzaro reaction, Clemmensen & Wolff-Kishner reductions, Grignard reagent
Unit 11: Organic Nitrogen Compounds	Amines (1°/2°/3°), Classification, Identification, Diazonium salts, Preparation & synthetic importance
Unit 12: Biomolecules	Carbohydrates (glucose, fructose, sucrose, starch, cellulose), Proteins (primary-quaternary structure), Nucleic acids (DNA/RNA), Vitamins

■ MATHEMATICS — 11 Units · 40 Questions · +1/0 Marking

Unit 1: Sets, Relations & Functions	Cartesian product, Union/Intersection, Equivalence relations, One-one/onto mappings, Composition of functions
Unit 2: Complex Numbers & Quadratic Equations	a+ib representation, Argand plane, Roots & coefficients, Nature of roots, Symmetric functions, Equations reducible to quadratic
Unit 3: Matrices, Determinants	Order 2×3 matrices, Minors, Cofactors, Area of triangle, Transpose, Symmetric/skew-symmetric, Inverse, System of linear equations, Rank method
Unit 4: Combinatorics	Permutations (without repetition, constraint), Combinations, P(n,r) and C(n,r), Simple applications
Unit 5: Algebra — Theory of Equations	Roots & coefficients relation, Reciprocal equations, Complex roots in conjugate pairs, Transformation of equations
Unit 6: Differential Calculus & Applications	Limits, Continuity, Differentiation up to 2nd order, Rate of change, Maxima/Minima, Tangents & normals, Rolle's & Lagrange's theorem, Differential equations
Unit 7: Integral Calculus & Applications	Standard integrals, Integration by substitution, Trigonometric identities, Properties of definite integrals
Unit 8: Analytical Geometry	Straight lines (all forms), Circles, Conics (Parabola/Ellipse/Hyperbola), 3D geometry (Direction cosines, Skew lines, Planes)
Unit 9: Vector Algebra	Addition, Scalar & vector products, Triple products, Application to plane geometry
Unit 10: Statistics & Probability	Mean, Median, Mode, SD, Variance, Bayes' theorem, Binomial, Poisson & Normal distributions
Unit 11: Trigonometry	Ratios, Compound angles, Identities, Inverse functions, Properties of triangles, Heights & distances

■ BIOLOGY — 10 Units · 40 Questions (PCB Stream) · +1/0 Marking

Unit 1: Diversity of Living World	5-Kingdom classification, Monera, Protista, Fungi, Plants (Algae→Angiosperm), Animals (Non-chordate to Class level), Binomial nomenclature
Unit 2: Structural Organisation in Plants & Animals	Plant tissues, Morphology (Root/Stem/Leaf/Flower/Fruit/Seed), Fabaceae/Solanaceae/Liliaceae families, Cockroach & Frog anatomy
Unit 3: Cell Structure & Function	Prokaryotic/Eukaryotic cells, Cell organelles, Biomolecules, Enzymes (types, properties, action), Cell cycle, Mitosis & Meiosis
Unit 4: Plant Physiology	Osmosis, Water potential, Transpiration, Mineral nutrition (macro/micro), Photosynthesis (C3/C4, light/dark reactions), Respiration (TCA cycle), Plant hormones
Unit 5: Human Physiology	Digestion, Breathing & respiration (exchange, transport), Circulation (ECG, double circulation), Excretion, Locomotion, Neural control, Endocrine system

Unit 6: Reproduction	Asexual & sexual reproduction, Flowering plant reproduction, Human reproduction (gametogenesis, menstrual cycle, embryo development), ART (IVF/ZIFT/GIFT)
Unit 7: Genetics & Evolution	Mendelian inheritance, Deviations (incomplete dominance, co-dominance, multiple alleles), Blood groups, Chromosomal disorders (Down's/Turner's/Klinefelter's), DNA replication, Central dogma, Hardy-Weinberg
Unit 8: Biology & Human Welfare	Diseases (Malaria, Dengue, COVID, Filariasis, Typhoid, AIDS), Immunology, Vaccines, Food production (plant breeding, tissue culture), Microbes in industry, Antibiotics
Unit 9: Biotechnology & Applications	Recombinant DNA technology, Bt crops, RNA interference, Human insulin, Gene therapy, Stem cells, DNA fingerprinting, Biosafety issues
Unit 10: Ecology & Environment	Ecosystems (structure, energy flow, nutrient cycles), Biodiversity (conservation, hotspots, Red Data Book), Air/Water pollution, Greenhouse effect, Ozone depletion

■ ENGLISH & APTITUDE — 20 Questions · +1/0 Marking

Aptitude: Data Interpretation	Tabular charts, Pie charts, Bar charts, Line graphs, Mixed charts
Aptitude: Data Sufficiency	All topics combined — determine if given data is sufficient to answer the question
Aptitude: Syllogism	2-3 statements with 2-4 conclusions — evaluate validity of conclusions
Aptitude: Number Series & Coding	Number series, Letter-to-letter coding, Letter-to-number coding, Mixed coding, Decoding
Aptitude: Clocks, Calendars & Directions	Mirror images of clock, Angle-based questions, Gain/loss per day, Finding days/dates, Directions & distance between two points
English: Reading Comprehension	Short passages & lines of poems — test comprehension, grammar, and pronunciation at higher secondary level